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April 26, 1977
SUPERSEDING
QQ-T-590B
July 30, 1968

FEDERAL SPECIFICATION

TOOL STEEL, HIGH-SPEED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tungsten-type and molybdenum-type high speed steels as described in ASTM A 600 which are used primarily in the fabrication of tools (see 6.1).

1.2 Classification. Material in accordance with this specification is classified by chemical composition. The types correspond to respective AISI designations.

1.2.1 Types T1, T2, T4, T5, T6, T8 and T15 are characterized by a controlled high tungsten content along with other alloying elements.

1.2.2 Types M1, M2, M3, M4, M6, M7, M10, M30, M33, M34, M36, M41, M42, M43, M44, M46, and M47 are characterized by a controlled high molybdenum content along with other alloying elements.

1.2.3 Types M2, M3, and M10 are further classified according to carbon range.

2. APPLICABLE DOCUMENTS

2.1 The applicable documents shown in ASTM A 600, Tool Steel, High-Speed, form a part of this specification. Unless otherwise specified in the contract or order, the latest revision of A 600 accepted by the Government shall apply.

3. REQUIREMENTS

3.1 The requirements for chemical composition, hardness, macrostructure and decarburization shall be as specified in A 600.

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3.2 Dimensions and dimensional tolerances shall be in accordance with the requirements of A 600, section 10 and tables 8 to 21 as applicable.

3.3 Workmanship. High-speed tool steels shall meet the requirements of section 11 of A 600.

3.4 When ultrasonic or cleanliness requirements and tests are specified in the contract or order, they shall be in accordance with the supplementary requirements of A 600.

3.5 Identification marking. The tool steel material shall be marked with the name or brand of manufacturer, number of the specification, heat number, grade or type, and where appropriate, the size, length, and weight. Unless otherwise specified in the contract or order, the method of marking is at the option of the manufacturer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order, a lot shall consist of all material submitted for inspection at the same time, of the same heat, type, finish, and size, and when heat treated, subjected to the same heat treatment.

4.3 Sampling.

4.3.1 For chemical analysis. At least one sample shall be selected from each heat to determine compliance with the requirements of table I of A 600.

4.3.2 For hardness test. Five samples or five percent (whichever is less) of the pieces in a lot shall be randomly selected for hardness tests.

4.3.3 For macrostructure examination. One sample shall be selected from each lot as described in A 600.

4.3.4 For decarburization examination. One sample shall be selected from each lot as described in A 600.

4.3.5 Examination, inspection, and test procedures for the high-speed tool steel material shall be conducted in accordance with A 600.

4.3.6 Rejection and retest shall be in accordance with sections 15 and 16 of A 600.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, loading, and marking shall be carried out as described in section 13 of A 600.

6. NOTES

6.1 Intended use. The tool steels covered by this specification are intended for the manufacture of tools used for machining materials at high speeds. In general it covers tools which are subject to severe duty.

6.1.1 Tungsten-base, high-speed tool steels. The classes in this group are characterized by high red hardness (resistance to heat) and wear resistance. They also have good depth of hardening and very high wear-resistance properties but have low toughness. T-15 is the most wear-resistant grade of this group. Cobalt additions increase the red hardness at a sacrifice in toughness. The presence of many wear-resistant carbides in a hard heat resistant matrix makes these steels suitable for cutting tool applications, while their toughness allows them to outperform the sintered carbides in delicate tools and interrupted-cut applications. Applications are almost entirely for such cutting tools as tool bits,drills, reamers, taps, broaches, milling cutters and hobs, although some usage is found for dies and punches and for high-load, high temperature structural components.

6.1.2 Molybdenum-base, high-speed tool steels. The classes in this group are very highly wear-resistant with deep hardening properties and fairly good nondeforming properties. These classes have low toughness and medium machinability properties. The molybdenum-base steels are somewhat more sensitive than the Tungsten base steels to hardening conditions, particularly to temperature and atmosphere, since they decarburize and overheat easily under adverse treating conditions. These classes are used for cutting tools of all kinds.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of tool steel.
c. When a revision other than the latest revision of the document is to be used (see 2.1).
d. When a particular method of marking is required (see 3.5).
e. Form, dimensional requirements, finish, condition, and supplementary requirements shown in section 4 of ASTM A 600.

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